MySQL Setup and Connection

This course uses **MySQL 8+** as the primary database. You can interact with MySQL through the **command-line client (CLI)** or **MySQL Workbench** (GUI).

Module Outcome:

- Install MySQL 8+ and open it from CLI or Workbench with a password you actually remember
- Connect, run `SELECT 1`, and know the service is up
- Keep root credentials off GitHub and off the projector

Installing MySQL

Windows

1. Download **MySQL Installer** from https://dev.mysql.com/downloads/installer/
2. Choose **MySQL Server** and **MySQL Workbench** during setup.
3. Set a **root password** during installation and note it down.
4. Complete the installer and start the MySQL service.

macOS

```
brew install mysql
brew services start mysql
mysql_secure_installation
```

Linux (Ubuntu/Debian)

```
sudo apt update
sudo apt install mysql-server
sudo mysql_secure_installation
```

Verify installation:

```
mysql --version
```

Expected output: `mysql Ver 8.x.x`

Installing MySQL Workbench

1. Download from https://dev.mysql.com/downloads/workbench/
2. Install and open MySQL Workbench.
3. Workbench provides a visual interface for writing queries, viewing schemas, and managing databases.

Connecting via CLI

Step 1: Open terminal (macOS/Linux) or Command Prompt (Windows).

Step 2: Connect as root:

```
mysql -u root -p
```

Enter your root password when prompted.

Step 3: Create a database for practice:

```
CREATE DATABASE training_db;
USE training_db;
```

Step 4: Verify connection:

```
SELECT DATABASE();
SHOW TABLES;
```

Step 5 (before querying homework): Load sample tables from `labs/training_db_seed.sql`. Module 02 still has you `CREATE player` yourself. How to run the script: `labs/README.md`.

Common CLI commands:

Command	Description
`mysql -u root -p`	Connect as root
`mysql -u username -p -h hostname`	Connect to remote server
`exit` or `quit`	Disconnect from MySQL
`SHOW DATABASES;`	List all databases
`USE database_name;`	Switch to a database

Connecting via MySQL Workbench

Step 1: Open MySQL Workbench.

Step 2: Click **+** next to "MySQL Connections" to add a new connection.

Step 3: Configure:

Field	Value
Connection Name	Local MySQL
Hostname	`127.0.0.1` or `localhost`
Port	`3306`
Username	`root`

Step 4: Click **Test Connection** → enter password → **OK**.

Step 5: Double-click the connection to open. Use the SQL editor tab to run queries.

Step 6: Select your database:

```
USE training_db;
```

Useful MySQL Commands (Both CLI and Workbench)

```
-- Show current database
SELECT DATABASE();

-- Show all tables in current database
SHOW TABLES;

-- Describe table structure
DESCRIBE player;
-- or
SHOW COLUMNS FROM player;

-- Show MySQL version
SELECT VERSION();
```

Creating a Practice User (Optional)

For training environments, create a dedicated user instead of using root:

```
CREATE USER 'trainee'@'localhost' IDENTIFIED BY 'password123';
GRANT ALL PRIVILEGES ON training_db.* TO 'trainee'@'localhost';
FLUSH PRIVILEGES;
```

Connect as the new user:

```
mysql -u trainee -p training_db
```

Troubleshooting

Problem	Solution
`Access denied for user`	Check username/password; verify user has privileges
`Can't connect to MySQL server`	Ensure MySQL service is running (`brew services start mysql` or Windows Services)
`Unknown database`	Run `CREATE DATABASE training_db;` first
Port 3306 in use	Check if another MySQL instance is running

Note: All SQL examples in this course assume you are connected to MySQL 8+ and have run `USE training_db;` (or your project database) before executing queries.

What next: MYSQL README — two teaching tracks, homework, and the seed lab.